Faculty of Science and Technology

Programme Structure & Syllabus

For

Second Year

Bachelor of Science (Honors/ Research)

Computer Science

(BSC CS)

P-2024

With Effect from Academic Year 2024– 2025

	Authority	Date
Proposed by	Board of Studies in Computer Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS **Dean of Faculty** **Director, IQAC**

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Science		
Emerge as renowned research and innovation-oriented department in the field of Information Technology		
Mission of the Department of Computer Science		
M1	To impart industry centric education through research and innovation	
M2	To develop skilled professionals to meet global socio-economic challenges	
M3	To introduce best practices in software development adhering to moral and ethics	
M4	To foster collaborative approach and leadership skills.	
Values of the Department of Computer Science		
Integrity		Brightness
Innovation		Diversity
Ethics		Competitive
Leadership		Lucidity

Mapping of Mission Statement of Department to University Mission Statement

Mission Statement	VU-M1	VU-M2	VU-M3	VU-M4
M1	2			
M2		3		
M3			3	
M4				2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	Graduates will have a thorough foundational knowledge of the principles and practices of Computer Science, and be proficient in applying them. (Based on Knowledge)
PEO2	To be able to adapt new tools/techniques to effectively contribute as an individual and lead the software development team to provide economically and technically viable solutions to a problem. (Based on Skills)
PEO3	Commit to ethical and moral practices for providing solutions to societal problems through life-long learning (Based on Attitude)

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3
M1	3	2	3
M2	1	3	2
M3	1	2	3
M4	1	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

1. Programme Outcomes (POs)

PO No.	Statement
PO1	Possess profound knowledge of theory and principles of basic mathematics, statistical techniques and algorithms in the field of computer science.
PO2	Ability to think critically and select economically and technically viable solutions to a given problem.
PO3	Ability to design, develop and deploy solutions for identified problems from diverse domains.
PO4	Provide proof of concept and present novel solutions to challenges encountered by industry.
PO5	Adopt new programming languages, tools, and processes followed in industry
PO6	To work independently or as an effective team member of a software development project.
PO7	Effectively communicate with diverse clients and stakeholders for software project management
PO8	Interpersonal skills to interact with people from diverse cultural backgrounds
PO9	Adhere to the professional, ethical and legal practices followed in the IT industry.
PO10	Possess software project management skills like: cost estimation, effort estimation, scheduling, and team building.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
PEO1	3	2	3	2	1	-	-	-	-	-
PEO2	1	3	-	2	3	2	2	1	1	2
PEO3	-	-	2	-	-	1	1	2	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	**Statement**
PSO1	Choose appropriate data structure, algorithm, and software development paradigm to address real life problems.
PSO2	Apply modern tools and techniques used in Data Science and Game Development.
PSO3	Provide the solutions to computing challenges from multidisciplinary fields.
PSO4	Possess a research and development attitude focusing on knowledge creation and innovation.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3
PEO1	3	1	2
PEO2	2	3	2
PEO3	1	2	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty	Science and Technology	**Pattern**	*2024*
Department	Computer Science	**Date (w.e.f.)**	*e.g. 15/07/2024*
Programme	B.Sc. – Computer Science		

Semester -III										
Course Type	**Course Code**	**Course Name**	**Teaching Scheme (Hours/Week)**			**Credit**	**Examination Scheme and Marks**		
			L	**T**	**P**	**C**	**CIE**	**ESE**	**Total**
DSC	BSCCS24301	Data Modeling and Visualization	2	1	0	3	60	40	100
DSC	BSCCS24302	Data Modeling and Visualization Lab	0	0	2	1	15	10	25
DSC	BSCCS24303	Data Structure	2	0	0	2	30	20	50
DSC	BSCCS24304	Data Structure Lab	0	0	2	1	15	10	25
CC	BSCCS24305	Statistical Methods	2	0	0	2	50	0	50
CEC	BSCCS24306	Technology Training of Society	0	0	4	2	50	0	50
MC	MT24CS03A	VU Level - PHP and MySQL	2	0	2	3	60	40	100
OE	OE24CS03A	VU Level - Green Computing	3	1	0	4	60	40	100
SEC	SE24CS03A	VU Level - Professional Writing	1	1	0	2	50	0	50
		Total	**12**	**3**	**10**	**20**	**390**	**160**	**550**
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Semester -IV										
Course Type	**Course Code**	**Course Name**	**Teaching Scheme (Hours/Week)**			**Credit**	**Examination Scheme and Marks**		
			L	**T**	**P**	**C**	**CIE**	**ESE**	**Total**
DSC	BSCCS24401	Object Oriented Programming	3	0	0	3	60	40	100
DSC	BSCCS24402	Object Oriented Programming Lab	0	0	2	1	15	10	25
DSC	BSCCS24403	Data Mining and Analytics	2	1	0	3	60	40	100
CC	BSCCS24404	Discrete Mathematics	2	0	0	2	50	0	50
FP	BSCCS24405	Industry Project	0	0	4	2	50	0	50
VSC	BSCCS24406	Data Science Using Python	1	0	2	2	50	0	50
MC	MT24CS04A	VU Level - Go Programming	2	0	2	3	60	40	100
AEC	AE24CS02A	VU Level- Modern Indian Language	3	0	2	4	60	40	100
Total			**13**	**1**	**12**	**20**	**405**	**170**	**575**
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Name of Head of Department

Sign:

Date:

Name of Dean

Sign:

Date:

BSCCS24303: Data Structure

Course Type	DSC	Semester	III

Teaching Scheme		Credits		Examination Scheme	
Lecture	02 Hr./Week	Lecture	2	CIE Marks	30
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	20
Practical/Studio	00 Hr./Week	Practical/Studio	0	Total Marks	50
Total	02 Hr./Week	Total	2		

Course Description

It teaches the ways of organizing in the form Stack, Queues, Arrays, Linked Lists and much more representation of the data and it covers the ways to perform searching or retrieving the data elements from the storage. It also emphasizes the importance of file and hashing concepts in data structures

Course Outcomes

CO No.	Statement
1	To understand the representation and applications of stack and Queue data structure.
2	To perform various operations on linked lists
3	To demonstrate the use of binary tree traversals and perform operations on them.
4	To Apply the Graph data structure to solve the applications of it.
5	To able to organize the data in files and further implement techniques for retrieving and strong the data

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Remember
CO2	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Understand
CO3	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Apply
CO4	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Analyze
CO5	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Evaluate
															Create

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Unit No 1	Stack & Queue Data Structure	6 Hours	CO	BTL
Sorting: Bubble Sort, Insertion Sort. Searching methods: linear search,binary search. Introduction to Data structures and design of algorithms, Abstract Data Type, ADT representations. Stacks: Fundamentals of stack, Representation and Implementation of stack using arrays,			1	1,2,3

Applications of stack: Decimal to Binary Conversion, reversing a string, Parsing: Well-formed parenthesis, Different expression conversions and evaluation, representation of multiple stacks. Queues: Fundamentals of queue, Representation and Implementation of queue using arrays, Circular queue: representation and implementation, Applications of queue: Josephus Problem, Job Scheduling, Queue Simulation, Categorizing Data, Doubly Ended Queue, Priority queue, representation of multiple queues.				
Unit No 2	**List Data Structure**	**6 Hours**		
Dynamic Memory allocation, Array representation using dynamic memory allocation, Concept of linked organization, singly linked list, doubly linked list, circular linked list, Insertion, Deletion and traversal on above data structures. Representation and manipulations of polynomials using linked lists. Implementation of linked Lists for Stacks and Queues, Generalized Linked List, operations on GLL like copy, Equality.			2	1,2,3
Unit No 3	**Tree Data Structure**	**6 Hours**		
Introduction to Tree data structures, Properties of Trees, Binary Trees, Applications of Binary trees, Binary Tree traversal, Expression trees and their usage, and Binary Search trees, Applications of Binary Trees.			3	1,2,3
Unit No 4	**Graph Data Structure**	**6 Hours**		
Introduction to Graphs, Types of graphs: Directed, Undirected, Weighted graph, Representation of graphs using adjacency matrix, adjacency list, Traversals: Depth First and Breadth First, Spanning Trees, Kruskal's and Prim's algorithms for minimum spanning tree.			4	1,2,3
Unit No 5	**File Structure & Hashing**	**6 Hours**		
File structures: Introduction to Files organization, Sequential Files (Structure, Operations, Disadvantages, Areas of use), Direct File Organization, Indexed Sequential File Organization. Accessing data from Files. Relation between data structures & Files. Hashing Techniques: Hash tables, Hash functions: Division, folding and Mid square methods, Collision Resolution Strategies: Linear Probing, rehashing, Open addressing and Chaining, Chaining with and without replacement, Chaining with linked list, Table Overflow, Rehashing, Open and closed hashing.			5	1,2,3

Textbooks

1	*Data Structures using C , C++, Andrew S. Tanenbaum, Yedidyah Langsam, Moshe Augenstein, Pearson/PHI, Second Edition, 1992, ISBN: 0876926960, 9780876926963.*

Reference Books / Journal Articles / Weblink

1	Schaum's Outline of Theory and Problems of Data Structures -Book by Seymour Lipschutz
2	Data Structures and Algorithm, A.V. Aho, J . E . Hopcroft, J . D . Ulman, Pearson Education
3	Data Structures, S. Sahni and E. Horowitz, Galgotia Publications

BSCCS24304 :: Data Structure Lab

Course Type	DSC	Semester	III

Teaching Scheme		Credits		Examination Scheme	
Practical/Studio	02 Hr./Week	Practical/Studio	1	CIE Marks	15
				ESE Marks	10
				Total Marks	25

Course Outcomes

CO No.	Statement
1	To understand the representation and applications of stack and Queue data structure.
2	To perform various operations on linked lists
3	To demonstrate the use of binary tree traversals and perform operations on them.
4	To Apply the Graph data structure to solve the applications of it.
5	To able to organize the data in files and further implement techniques for retrieving and strong the data

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Remember
CO2	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Understand
CO3	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Apply
CO4	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Analyze
CO5	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Evaluate
															Create

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1.	Program based on searching and sorting methods.		
2.	Program for different Stack applications	1	3
3.	Program for Expression conversion and evaluation using Stack	1	3
4.	Program for different Queue applications	1	3
5.	Program for different types of Linked list implementation Create a linked list with nodes having information about a student and perform a. Insert a new node at specified position. b. Delete of a node with the roll number of student specified. c. Reversal of that linked list.	2	3

	Create doubly linked list with nodes having information about an employee and perform Insertion at front of doubly linked list and perform deletion at end of that doubly linked list.		
6.	Program on Binary and Binary Search tree and its traversals	3	3
7.	Program on BST operations	3	3
8.	Program on Graph traversals (BFS and DFS)	4	3
9.	Program on min. spanning tree and Dijkstra's algorithm	4	3
10.	Program on Collision handling techniques.	5	3
11.	Program on Sequential file creation and accessing	5	3

Textbooks

1	*Data Structures using C , C++, Andrew S. Tanenbaum, Yedidyah Langsam, Moshe Augenstein, Pearson/PHI, Second Edition, 1992, ISBN: 0876926960, 9780876926963.*

Reference Books / Journal Articles / Weblink

1	Schaum's Outline of Theory and Problems of Data Structures -Book by Seymour Lipschutz
2	Data Structures and Algorithm, A.V. Aho, J . E . Hopcroft, J . D . Ulman, Pearson Education
3	Data Structures, S. Sahni and E. Horowitz, Galgotia Publications

BSCCS24305: Statistical Methods

Course Type	CC	Semester	III

Teaching Scheme		Credits		Examination Scheme	
Lecture	02 Hr./Week	Lecture		CIE Marks	50
Tutorial	0 Hr./Week	Tutorial		ESE Marks	00
Practical/Studio	0 Hr./Week	Practical/Studio		Total Marks	50
Total	3 Hr./Week	Total			

Course Description

The purpose of this course is to familiarize students with basics of Statistics. This will be essential for prospective researchers and professionals to know these basics.

Course Outcomes

CO No.	Statement
1	Understand the Statistical Concepts Population and Sample, Variable and Attributes
2	Apply Knowledge in Professional Software for Results Interpretation using Statistical data
3	Develop aligned representation using the Statistical data
4	Demonstrate their knowledge of the basics of inferential statistics by making valid generalizations fr sample data
5	Structure and Solve the Problems using Mathematical representations and Relationships

Mapping of COs to POs and PSOs

CO Number	POs									PSOs			BTL		
	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO7	PO 8	PO 9	PSO 1	PSO 2	..			
CO1	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	Remember
CO2	**-**	**2**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**10**	**-**	**2**	**-**	**4**	Understand
CO3	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	Apply
CO4	**1**	**2**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**4**	Analyze
..	**-**	**2**	**3**	**4**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**3**	**-**	Evaluate
															Create

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No – I	**Title of Unit - Basic Concepts**	**Hours – 09**	**CO**	**BTL**
Definition of Statistics, concepts of statistical population and sample, parameters and statistics, statistical data analysis. Data: quantitative and qualitative, variables, attributes. Scales of measurement: nominal, ordinal, interval and ratio. Frequency distribution (inclusive and exclusive), cumulative frequency distribution.			**1**	**2**
Unit No – II	**Title of Unit – Presentation of data**	**Hours – 09**	**2**	**1**
Presentation: Graphical including histogram, frequency polygon, bar-chart, box-plot. Measures of central tendency. Partition values: Quartiles, deciles and percentiles, partition values using graphical representation.				
Unit No – III	**Title of Unit - Measures of Dispersion**	**Hours – 09**	**3**	**4**
Range, quartile deviation, coefficient of quartile deviation, mean deviation, standard deviation, variance, coefficient of variation. (For raw data, ungrouped frequency distribution, exclusive type frequency distribution)				
Unit No – IV	**Title of Unit – Moments**	**Hours – 09**	**4**	**3**
Raw and Central moments: Definition, computations for ungrouped and grouped data (only up to first our moments). Relation between raw and central moments up to fourth order. Skewness and Kurtosis.				
Unit No – V	**Title of Unit – Correlation and Regression**	**Hours – 09**	**5**	**5**
Definition, scatter diagram, Karl Pearson‟s correlation, rank correlation and its limit. Linear regression, regression coefficients and its properties. Principle of least squares, fitting of polynomials and exponential curves.				

Textbooks

1	Gupta, S.C. and Kapoor, V.K. (2002) : Fundamentals of Mathematical Statistics, S. Chand and Sons, New
2	Goon A.M., Gupta M.K. and Dasgupta B. (2002): Fundamentals of Statistics, Vol. I & II, (8th Edn). The Press, Kolkata.
3	Kulkarni, M.B., Ghatpande, S.B. and Gore, S.D. (1999): Common statistical tests. Satyajeet Prakashan, Pu

Reference Books / Journal Articles / Weblink

1	Business Statistics, By Naval Bajpai
2	Mukhopadhyay, P. (1996): Mathematical Statistics. New Central BookAgency, Calcutta, Introduction to Mathematical Statistics, Ed. 4 (1989),MacMillan Publishing Co. New York.
3	Ross, S.M. (2006): A First course in probability. 6th Ed^{n} Pearson

BSCCS24306: Technology Training of society

Course Type	CES	Semester	III

Teaching Scheme		**Credits**		**Examination Scheme**	
Lecture	00 Hr./Week	Lecture	0	CIE Marks	50
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	00
Practical/Studio	04 Hr./Week	Practical/Studio	4	Total Marks	50
Total	30 Hr./Week	Total			

Course Description

This course explores the multifaceted relationship between technology and society, with a specific focus on the design, implementation, and evaluation of technology training programs. Students will examine the socio-cultural implications of technology adoption, analyze effective pedagogical strategies for technology training, and develop practical skills for designing inclusive and accessible training initiatives. Through a combination of theoretical discussions, case studies, and hands-on projects, students will gain insights into the role of technology in shaping individual behaviors, community dynamics, and societal structures.

Course Outcomes

CO No.	Statement
1	Summarize the concepts of programming before actually starting to write new programs
2	Comprehend what happens in the background when the programs are executed
3	Develop logic for Problem Solving.
4	Develop basic constructs of programming such as data, operations, conditions, loops, functions etc.
5	Apply the problem-solving skills using syntactically simple language i. e. C programming language

Mapping of COs to POs and PSOs

CO Number	POs									PSOs			BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	..	..	PSO1	PSO2	..	
CO1	3	1	2							3	2		1,2
CO2		2	3	1						2	1		2,4
CO3				3	2	1				2	1		1,3,4
CO4			2	3	1					1	3		5,6
..													

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom’s Taxonomy Level

Course Content

Unit No I	Introduction to Technology and Society and Ethical Considerations in Technology	8 Hours	CO	BTL
Session 1: Overview of Technology and Society Introduction to key concepts and theories, Historical perspectives on technological revolutions Discussion on the digital divide and its implications, Activities: - Group discussion on current technological trends, - Case study analysis of societal impacts of past technological revolutions Ethical Considerations in Technology: Session 2: Ethical implications of technology adoption and deployment, Case studies on ethical dilemmas in technology training, Activities-: Ethical decision-making exercises related to technology use, Role-playing scenarios on ethical technology deployment			1,2	1,2
Unit No II	**Socio-Cultural Impact of Technology and Designing Technology Training Programs**	**8 Hours**		
Session 3: Analysis of Technology's Impact on Society, Study of technology's influence on individual behaviors and social interactions, Examination of cultural dimensions of technology usage, Activities-: Conducting surveys or interviews on technology usage patterns Case studies on cultural adaptations to technology in different societies Designing Technology Training Programs: Session 4: Principles of inclusive design and accessibility in technology training, Hands-on exercise: Designing accessible learning materials, Activities-: Group project: Designing accessible technology training modules, Workshop on inclusive design principles			2,3	2,4
Unit No. III	**Technology Training for Community and Workforce Development**	**8 Hours**		
Session 5: Technology and Workforce Development, Analysis of the changing nature of work in the digital age, Strategies for upskilling and reskilling the workforce, Activities-: Role-playing scenarios on workforce training challenges, Case study analysis of successful workforce development initiatives Session 6: Leveraging technology for community empowerment and social innovation, Design thinking methodologies for co-creating technology solutions, Activities-: Design thinking workshop: Developing community-driven tech solutions, Case study discussion on community-driven technology projects			3,4	1,3,4
Unit No IV	**Evaluation and Future Trends**	**6 Hours**		
Session 7: Evaluation of Technology Training Programs, Designing evaluation frameworks for measuring impact, Formative and summative assessment strategies, Activities-: Workshop: Developing rubrics for assessing technology competency, Peer-review of technology training program evaluations			2,4	2,4,5

Textbooks

1	Technology and Society Issues for the 21st Century and Beyond, by Linda S. Hjorth and Barbara A. Eichler e-book
2	Digital Literacy for Dummies, by Faithe Wempen
3	Ethics for the Information Age, by Michael J. Quinn

Reference Books / Journal Articles / Weblink

1	Academic journals and articles on technology and society
2	Case studies from reputable sources such as MIT Technology Review, Harvard Business Review, etc.

MT24CS03A :: PHP and MySQL

Course Type	PC	Semester	IV

Teaching Scheme		Credits		Examination Scheme	
Lecture	02 Hr./Week	Lecture	2	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	02 Hr./Week	Practical/Studio	1	Total Marks	100
Total	04 Hr./Week	Total	3		

Course Description

PHP is a widely used programming language which works on the principal of server-side scripting to produce dynamic Web pages. To introduce how PHP can be combined with MySQL to integrate database functions into Websites.

Course Outcomes

CO No.	Statement
1	To understand basics of PHP and to implement PHP script using Decisions and Loops
2	To develop PHP applications using Strings, Arrays and Functions.
3	Use HTML form elements that work with any server-side language.
4	To display and insert data using PHP and MySQL

Mapping of COs to POs and PSOs

CO Number	Pos										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		1	1		3										1, 2,
CO2	3	2									2				1, 4
CO3	3		3								2	3			1, 3, 5
CO4	3		1	2							2	3	2	2	1, 3, 5
CO5		3	2	2							2		3	3	1, 3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom’s Taxonomy Level

Course Content

Unit No: 1	**Title of Unit: Introduction to PHP**	**4 Hours**	**CO**	**BTL**
Evaluation of Php, Basic Syntax, Defining variable and constant, Php Data type, Operator and Expression.			1	1, 2
Unit No: 2	**Title of Unit: Decisions and loop**	**6 Hours**	**1**	**2, 4**
Making Decisions, Doing Repetitive task with looping, Mixing Decisions and looping with Html. Understanding Exception and error, Try, catch, throw.				
Unit No: 3	**Title of Unit: Function & Arrays**	**8 Hours**	**2**	**3, 4**
What is a function, Define a function, Call by value and Call by reference, Recursive function, String Creating and accessing, String Related Library function Anatomy of an Array, Creating index based and Associative array Accessing array, Element Looping with Index based array, Looping with associative array using each () and foreach(), Some useful Library function.				
Unit No: 4	**Title of Unit: Handling Html Form with Php**	**8 Hours**	**3**	**2, 3**
Capturing Form, Data Dealing with Multi-value filed, and Generating File uploaded form, redirecting a form after submission.				
Unit No: 5	**Title of Unit: Database Connectivity with MySql**	**4 Hours**	**4**	**1, 2,3**
Introduction to RDBMS, Connection with MySql Database, Performing basic database operation(DML) (Insert, Delete, Update, Select), Setting query parameter, Executing query.				
Program List				
1. Write a PHP program to print sum of digits. 2. Write a PHP program to check even odd number. 3. Write a PHP program to check prime number. 4. Write a PHP program to reverse given string. 5. Write a PHP program to reverse given string. 6. Write a PHP program to calculate factorial of a number using function. 7. Write a program to insert / update/ delete data in MySQL via PHP script.				

Textbooks

1	PHP for Beginners, SPD publication
2	Robin Nixon, “Learning PHP, Mysql and Javascript with JQuery, CSS & HTML5”, O'REILLY, ISBN: 13:978-93-5213-015-3

Reference Books / Journal Articles / Weblink

1	www.php.net.in 11.
2	https://www.w3schools.com/php/php_mysql_intro.asp

OE24CS03A: Green computing

Course Type	Theory	Semester	3

Teaching Scheme		**Credits**		**Examination Scheme**	
Lecture	3 Hrs./Week	Lecture	3	CIE Marks	60
Tutorial	1 Hrs./Week	Tutorial	1	ESE Marks	40
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	100
Total	4 Hrs./Week	Total	4		

Course Objectives

This course enables students to minimize energy consumption, waste, and other environmental impacts of IT systems while lowering life cycle costs, thereby enhancing competitive advantage.

Course Outcomes

CO No.	**At the end of this course students will be able to …..**
1	To learn the fundamentals of Green Computing.
2	To understand the issues related with Green compliance, why to go green, toxins, power consumption and heat
3	To understand Current initiatives and standards for Green Computing , Solving the E-waste Problem, (StEP) principles of Globalinitiatives,
4	To identify and apply strategies to improve energy efficiency in data centers, hardware, and software.
5	To study and develop various case studies based on Green Computing

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		1	1		1								1		1	1	1,2
CO2		1	1		2								1		1	1	2,3
CO3		1	1		2								1		1	1	2,3
CO4		1	1		2								1		1	1	3,4
CO5													1				

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.01	**Introduction to Green Computing**	**9 hours**	**CO**	**BTL**
Definition and Scope of Green Computing , Environmental Impact of IT, Carbon footprint of computing devices and infrastructure, Energy Efficiency in IT, Sustainable Hardware and Software Design			1	1,2
Unit No.02	**Issues related to IT and Current Initiatives and Standards**	**9 Hours**		
An overview and Issues related to it Toxins, Energy Consumption and Efficiency Carbon Footprint Heat How can the issues be solved E-Waste Management Solving the E-waste Problem and Global Initiatives, STEP five principles			2, 3	3,4
Unit No.03	**Energy Efficiency and Techniques for improving energy efficiency**	**9 Hours**		
Introduction to Energy Efficiency Power Problems, Monitoring Power Usage, Servers Virtualization and Storage, Cooling, Calculating Cooling Needs Reducing Cooling Costs 3.7 Strategies for Cooling, Green Supply Chain			4	2,3,4
Unit No.04	**Strategies To Improve Energy Efficiency**	**9 Hours**		
Changing the way of work, Recycling, Define paper problem, Going Paperless, Designing Energy-Efficient Data Centers, Green Data Center Infrastructure: Trends and Technologies, Green Manufacturing Practices in IT Hardware, storage and Networking, Green Software Engineering Practices, Energy-Efficient Software Solutions			4	3,4,5
Unit No.05	**Case Studies Green Computing**	**9 Hours**		
Green Information Systems, Green Enterprise Architecture, Applying Green IT Strategies and Applications to a Home, Hospital, IT organizations, Adobe's Green Office, Google's Data Center Innovations, Microsoft's Carbon Negative Initiative, Apple's Commitment to 100% Recycled Materials, Amazon Web Services (AWS) and Renewable Energy			5	4,5,6

Textbooks

1	Green Computing: Tools and Techniques for Saving Energy, Money, and Resources By by Bud E. Smith CRC Press
2	Green Computing and Its Applications by Alok Kumar Singh Kushwaha, PhD (Editor) , Raj Kumar Patra, PhD (Editor) , Rohit Raja, PhD (Editor) , Sanjay Kumar, PhD (Editor) , Saurabh Kumar, PhD (Editor)

Reference Books / Journal Articles / Weblink

Reference Books	
1	Green IT: Reduce Your Information System's Environmental Impact While Adding to the Bottom Line By Toby Velte (Author), Anthony Velte (Author), Robert Elsenpeter (Author) publisher Tata McGraw -Hill
2	www.researchgate.com
3	https://www.techtarget.com

List of Tutorials

Sr. No	**Name of Tutorial**	**CO**	**BTL**
1	Designing Energy-Efficient Data Centers		
2	Implementing Green IT Policies in **Home, Hospital, IT organizations**		
3	E-Waste Management and Recycling		
4	Green Networking Solutions		
5	Sustainable Supply Chain Management in IT		
6	Energy-Efficient Cloud Computing		
7	Green IT Auditing and Reporting		

SE24CS03A :: Professional Writing

Course Type	SEC	Semester	III

Teaching Scheme		Credits		Examination Scheme	
Lecture	1 Hr./Week	Lecture	1	CIE Marks	50
Tutorial	1 Hr./Week	Tutorial	1	ESE Marks	0
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	50
Total	2 Hr./Week	Total	2		

Course Description

Today writing has become an important part of communicating ideas, concept, information and keeping the track of work. As the students in their graduate level prepare to interact with the professional world, professional writing becomes an important part so that students can communicate to their institutions, corporate or government sectors whether it would be writing a letter, an email, a newsletter, grant or proposal writing, memo writing, review paper or content writing, etc. In short, this course teaches a student to interact or communicate professionally in written form.

Course Outcomes

CO No.	Statement
1	Enumerate the concept of Professional Writing.
2	Summarize the types of Professional Writing.
3	Apply the applications of Professional Writing.
4	Analyse the errors and infer the use of Emergent Writing.
5	Assess the tools helpful for Professional Writing.
6	Prepare a case study on Professional Writing.

Mapping of COs to POs and PSOs

CO No.	Pos										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4	
CO1				1		1	3	3	3	1					1,2
CO2		3	2	1		1	3	3	3			2			2,4
CO3		3	3	2		2	3	3	3			3			1,3,4
CO4			3	2		3	3	3	3	2		1			5,6
CO5		3	3	3	3	3	3	3	3	2		3			
CO6			3	3		3	3	3	3	3		3	3		

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No. 1	**Beginning with Professional Writing**	**5 Hours**	**CO**	**BTL**
Introduction to Professional Writing; Communication skill in Professional Writing; The seven C's of Professional Writing; Grammar in Professional writing.			1	1,2
Unit No. 2	**Understanding the Types**	**6 Hours**		
Letters; Newsletters; Memos; Press Releases, Meeting Agenda and Minutes, Resume Writing – Importance, Structure, Do's and Don'ts; Cover Letter – Do's and Don'ts; Project Proposal Writing; Emails; Professional Journals or Websites; Handbooks,			2	1,2
Unit No. 3	**Applications**	**7 Hours**		
Business Communication – Email and Memos, Reports and Proposals; Technical Writing-Review Papers, Grants, Letter of Recommendation, Personnel Essays; Marketing and Public Relations – Content Creation, Press Releases, Advertising Copy, Newsletters; Grant and Proposal Writing – Grant Proposal, Business Plans			3	1,2,3
Unit No. 4	**Analysis**	**3 Hours**		
Grammatical Errors of ESL Learners; Project Based Learning in Writing Pedagogy; Emergent Writing			4	1,2,4
Unit No. 5	**Useful Tools**	**5 Hours**		
Word Processing Software; Editing Tools; Project Management Tools; Content Management Systems, Data Visualization and Presentation Tools; Document and Code Writing Tools.			5	1,2,4,5
Unit No. 6	**Writing a Professional Document**	**4 Hours**	**6**	**1,2,5,6**
Technical Writing for Software Documentation-API Documentation, Grant Proposal Writing for Patent; Annual Report in Business Communication; Marketing Copywrite in Digital Advertising; Investigative Journalism Report				

Tutorial List:

1. Difference between Professional and Academic Writing.
2. Communication skill used in Professional Writing.
3. The Seven C's of Professional Writing.
4. Write an Email explaining the summary of Project.
5. Give a Memo in written.
6. Create your professional Resume.
7. Write a Blog.
8. Write a Research Proposal

9. Write a Grant Proposal
10. Write a newsletter.

Textbooks

1	*"Professional Writing Skills in English", Abishith G. Rao, Dr. Binoy Mathew, Somasundar B.M., Clever Fox Publishing, 2023.*
2	"A Handbook of Letter Writing", S. C. Gupta, Arihant Publications, Third Edition, 2016.
3.	*"Basics of Writing Project Proposal", D. Kushwaha, New Delhi Publishers, 2013.*
4.	*"Technical Communication", Mike Markel; Stuart Selber, macmillan learning, Thirteenth Edition, 2021.*
5.	***"The Business Writer's Handbook", Alred, Gerald J; Brusaw, Charles T; Oliu, Walter E, Boston : Bedford/St. Martin's, Eighth Edition. 2006.***

Reference Books / Journal Articles / Weblink

1	*https://pressbooks.bccampus.ca/communicating/front-matter/introduction/ (accessed on 15/06/2024)*
2	*https://www.tcsion.com/courses/interview-and-job-prep/write-effective-resume-and-cover-letter/(accessed on 15/06/2024)*
3	*https://www.coursera.org/learn/business-writing (accessed on 15/06/2024)*
4	*https://www.coursera.org/learn/sciwrite (accessed on 15/06/2024)*
5	*https://www.coursera.org/learn/writing-for-children (accessed on 15/06/2024)*
6	*https://www.classcentral.com/course/youtube-grant-writing-61754 (accessed on 15/06/2024)*
7	*Rizwan, Msm. (2023). Grammatical Errors of ESL Learners: A Disruption in Professional Writing. International Journal of Research Publication and Reviews. 4. 2840-2843. 10.55248/gengpi.4.1023.102824.*
8	*Farihah, Nurul & Ajleaa, Nurul & Yingsoon, Goh & Azman, Nabila. (2024). FROM IDEAS TO ESSAYS: EVALUATING PROJECT-BASED LEARNING IN WRITING PEDAGOGY. Journal of Islamic Social Economics and Development. 128-1755. 10.55573/JISED.096222.*
9	*Debaryshe, Barbara. (2023). Supporting Emergent Writing in Preschool Classrooms: Results of a Professional Development Program. Education Sciences. 13. 961. 10.3390/educsci13090961.*